Graphic Equalisation — Cosmology Grounding

Horizons, Measures, and What Has No Outside

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Abstract

This companion develops the cosmological groundings of the substrate, and its work is mostly negative in a specific and useful way: it identifies which cosmological questions are ill-posed rather than merely hard, and says why on the account's own results. The spine is the measure problem of eternal inflation, which is the no-prior-free proposition in another domain — a probability over pocket universes requires a frame-free denominator, and the companion proves no such thing is a member of the space, so the difficulty is not a missing regularisation but a request that cannot be met. Around that: the universe *as a whole* is one of the three degenerate readings the base paper already prices at zero, so the observable universe is the object rather than a sample of one; a particle horizon is an instantiated boundary relation, and what lies beyond it is absence in the account's first-class sense rather than concealment, which permits modelling and refuses existence claims; and the cosmological principle divides cleanly into an observation that survives and an inference that does not, the second leg being modal reasoning of exactly the kind the base paper declines. *This paper derives no cosmological parameter, predicts no observation, and settles nothing empirical.* It says which questions have answers and which have none, which is a smaller thing, and it agrees with the field's own instincts more often than it corrects them — homogeneity is called an assumption and the measure problem is called unsolved, and both are correct in kind for reasons given here.

1 Scope, and what this adds

The physics companion already treats the initial state. It gives the Big Bang as a chosen entry rather than an intrinsic edge, the Past Hypothesis as a seed and therefore rightly an assumption, and fine-tuning inverted by self-location. *None of that is repeated here.* What follows takes up the questions that are cosmological rather than initial: what the subject of cosmology is, what a horizon is, what may be said about what lies past one, and what happens to probability when the sample space has no outside.

Two disciplines carry over and are stated once. The account prices operations and does not license them, and it distinguishes what is *observed* from what is *modelled* — a distinction the base paper makes for its own object and which does most of the work below.

2 The universe is not an object

Cosmology's ordinary subject does not exist, and the base paper already says so. That paper gives three readings which each yield nothing: a single relatum with nothing to relate to, the empty case, and *everything at once read from outside, which has nothing left over to stand in relation to it.* The universe as a whole is the third.

Proposition 2.1 (The totality is a degenerate reading)**.** *There is no frame from which everything is the relatum, since a frame is an instantiated boundary relation and a boundary requires something on the other side. So* the universe as a whole *names no object: what it names is a reading taken from a standpoint that does not exist, and such a reading returns nothing.*

The consequence is a correction and not a limitation. It is usual to say that we observe only part of the universe and must infer the rest. On this account the observable universe is not a sample of the object — *it is the object*, being what stands in relation to the frame doing the cosmology. The horizon does not truncate a larger thing; it individuates the thing there is.

This is less revisionary than it sounds, and the practice already runs this way. Cosmological parameters are fitted to what is inside the horizon, and the fits do not degrade for want of what is outside. A quantity that made no difference to any fit was doing no work in the account, which is the base paper's own test.

3 A horizon is a boundary relation

A particle horizon is the boundary relation itself, not a fence inside a larger space. The companion's frame is an instantiated boundary $\mathcal{B}_{W:G}$, and individuation is performed by the container rather than by the thing. A horizon individuates a cosmological frame in exactly that sense, and its dependence on the observer's worldline is then expected rather than awkward: different worldlines instantiate different boundaries, so they have different frames, and there is no frame containing all of them from which to say which is right.

Proposition 3.1 (Beyond a horizon is absence, not concealment)**.** *What stands in no relation to a frame is absent from it in the account's first-class sense, and absence is not a hidden presence.* So a claim that structure exists beyond the horizon is a claim about relata standing in no relation to the claimant*, which the base paper prohibits as an existence claim and permits as a model, on the same terms it permits its own object to be modelled and not observed.*

The permission matters as much as the prohibition. Nothing here forbids the standard extrapolation — that the universe continues past the horizon much as it does inside — and that extrapolation is useful, disciplined, and routinely made. *What is refused is the upgrade from model to fact*, and the refusal costs the practice nothing, because no measurement discriminates the two.

Remark 3.2 (Where this could be wrong, and it is testable)*. This is not idle.* Searches for signatures of bubble collisions in the microwave background are searches for a relation to something outside the horizon — and a detection would establish exactly the relation whose absence is assumed here. *The account does not predict the null.* It says that absent such a relation, what lies beyond is modelled and not observed; a found relation would make the outside a relatum, at which point it is inside the frame and the account applies to it as to anything else. *The question is empirical and the account does not prejudge it.*

4 The cosmological principle divides

Isotropy is observed. Homogeneity is inferred, and the inference has a modal leg. What is measured is that the sky looks the same in every direction to high precision. What is wanted is

that the universe is the same everywhere, and the step between them is the Copernican principle: that our position is not special among positions.

Proposition 4.1 (The second leg is modal and is declined)**.** Our position is not special *compares this position against positions we do not occupy. The base paper's commitment is that what bears on a claim is a demonstration and not an imagining, since imagining a world costs nothing and settles nothing.* So the inference from isotropy to homogeneity beyond the horizon is not available on this account*, and what survives is isotropy, which is measured, together with homogeneity within the horizon, which is also measured.*

The field's own vocabulary already concedes this. Homogeneity is called an assumption, not a result, and it is called that by people with no interest in this account. *What is added here is not the suspicion but the reason*: it is not an assumption because the measurement is hard, but because the quantity it asserts stands in no relation to the frame asserting it.

5 The measure problem is the no-prior-free proposition

This is the case where the account does real work, and it is the case cosmology already knows is broken.

Eternal inflation produces unboundedly many pocket universes. To extract a prediction — what a typical observer sees, how likely a given constant is — one needs a probability measure over them. *The theory does not supply one.* Different regularisations give different answers and some give absurd ones, and the difficulty is named and open.

Proposition 5.1 (The measure problem is ill-posed, not unsolved)**.** *A measure over all pockets requires a standpoint from which all of them are counted, and by the companion's no-prior-free proposition the zero of such an axis is not a member of the space: a frame standing in no relation to what it counts draws no distinction and returns nothing.* So the quantity being sought does not exist to be approximated. *The regularisations disagree because they are supplying, by stipulation, a denominator the theory cannot supply — and by the Narrativium companion's distinction a stipulated conversion carries exactly the authority of whoever stipulated it, which is why the answers track the stipulation rather than the physics.*

The consequence is the same one the adaptability companion reaches for intelligence. The difficulty is not that the measurement is hard but that the control it needs is not constructible, and an account that kept looking for a better regularisation would be searching for a member of an empty set. *This is a dissolution and not a solution*, and it predicts that the disagreement between measures will not converge — which is a claim about the literature's future rather than about the world, and is checkable.

Remark 5.2 (What this does not do to inflation)*. Nothing here bears on inflation as physics.* Inflationary models make horizon-scale predictions that are tested inside a frame and stand or fall on those. What is refused is a specific inferential move — probability over an ensemble with no outside — and a theory can be entirely correct while one inference drawn from it is malformed. *Conflating the two would be the error this paper is otherwise about.*

6 Anthropic reasoning inherits the same defect

Typicality arguments are the measure problem wearing a different hat. That we should expect to find ourselves in a typical position among observers requires a distribution over observers, which requires the counting standpoint just refused. *So the anthropic inference is not weak evidence but no evidence*, and this is a stronger statement than the usual complaint that the reference class is hard to choose.

The physics companion reaches the same place from the other direction and the two should be read together: there, fine-tuning is dissolved by self-location, since a different seed is a different observer and there is no trans-seed standpoint from which to compare. Here the same absence removes the distribution. *One prohibition, met twice.*

7 Two measurements that disagree

A word on the Hubble tension, and it is a word of caution rather than a result. Early-universe and late-universe determinations of the expansion rate disagree beyond their stated errors.

The account says only what it says. By the adaptability companion's result a measurement is a property of a four-place arrangement, and two determinations made under different arrangements are two measurements rather than one measurement made twice — so comparing them requires an argument that the moved coordinate does not bear. *That is exactly what the field is doing*, at length and carefully, which is why this section claims nothing further. *The account does not dissolve the tension, does not favour either value, and does not predict which coordinate is at fault.* It says the comparison is the work, which cosmologists established without help.

8 What is not claimed

No cosmological parameter is derived here, and none could be. No observation is predicted. Nothing above distinguishes between inflationary models, settles the expansion tension, or bears on dark energy's equation of state.

Nor is any of the negative work a criticism of practice. Homogeneity is already called an assumption; the measure problem is already called unsolved; the tension is already treated as a comparison problem. *The account agrees with the field's instincts and supplies reasons for them*, which is worth less than a prediction and more than nothing.

9 Falsifiability commitments

A relation to the outside. A confirmed signature of a bubble collision, or of any structure causally beyond the particle horizon, would establish a relation to what this paper treats as absent. That would not refute the account — the outside would become a relatum and the frame would enlarge — but it would refute this paper's application of it, which assumes no such relation obtains.

Convergence of measures. If the regularisations of eternal inflation converge on one measure by an argument internal to the theory, requiring no stipulation from outside, then Proposition 5.1 is wrong and the problem was hard rather than ill-posed.

Anisotropy that survives. Isotropy is the leg this paper keeps. A large-scale anisotropy that survives scrutiny would remove the observed half of the cosmological principle, and this paper would have kept the wrong leg.

Derivation of a seed. If a cosmological initial condition is ever derived rather than installed — from within, with no boundary condition imported — then the physics companion's seed result and this paper's dependence on it both fail.